

A Case Study on Repairability and Sustainability of AirPods

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Introduction

In the fast-paced world of consumer electronics, innovation often (unfortunately) races ahead of sustainability. Aesthetics, compactness, and performance generally dominate product briefs, while sustainability and repairability often are neglected. But at what ethical cost? This case study investigates the ethical dilemma presented by the specific design of Apple's AirPods: a product line that, while technically very fascinating and sophisticated, as well as globally popular, exemplifies troubling trends in the disposability of modern technology. Through the lens of repairability and sustainability, this paper argues that engineering decisions made under the guise of "aesthetics" and "technical progress" can sometimes be dishonest and nonbeneficial, while conflicting deeply with ethical obligations to consumers and the planet.

This particular case is not merely a fringe issue: it's a bellwether. AirPods are, in many ways, indicative of where the modern electronics industry is headed: wireless, sealed shut, and destined for the landfill. As engineers (and society at large) grapple with the indisputable climate crisis, scarce resources, and (hopefully) a necessary paradigm shift towards sustainability, this case exemplifies an urgent ethical challenge in real-world engineering practice.

Background

First introduced in Late 2016, Apple's AirPods redefined the wireless earbud market. According to an article from *Business of Apps*¹, by the end of 2024 (and across 4 generations of the product), over 432 million units had been sold. Despite their ubiquity and innovation, AirPods also became notorious for being essentially unrepairable. Teardowns conducted by *iFixit*² and others revealed that the lithium-ion batteries within each earbud and charging case are glued in with strong adhesives and encased in plastic shells that must be cracked and broken open to access internal components for repair. To put this bluntly, even skilled technicians can't fix them without destroying them.

¹ Business of Apps. "Apple Statistics (2025)." Business of Apps, 2025, <https://www.businessofapps.com/data/apple-statistics/>.

² iFixit. "AirPods Teardown." <https://www.ifixit.com/Teardown/AirPods+Teardown/75578>

The lifespan of an AirPods battery under regular use is roughly two to three years³. After that, the earbuds still function, but they no longer hold a charge that is satisfactory for most users. Therein lies the primary issue: users are encouraged to simply buy a new pair, rather than repair the old pair. Once those batteries degrade, the AirPods and all other functioning parts in them are now e-waste.

It seems Apple's rationale leans on technical necessity: that they could not achieve this level of performance and miniaturization if the product were designed to be opened and serviced. However, environmentalists and consumer rights advocates have pushed back against intentionally neglecting repairability. This approach undermines both sustainability and user autonomy, which are two increasingly important values in ethical engineering discourse.

Key Stakeholders

To understand the ethical weight of this case, we must consider the myriads of actors whose interests and responsibilities intersect within it. At the center, we have Apple: a company that has long built its brand on the intersection of cutting-edge innovation and seamless user experience. Its marketing materials are saturated with green symbolism and promises of carbon neutrality, yet its design decisions often obfuscate their true stance or sometimes undercut those very values. In the case of AirPods, Apple is both a protagonist and antagonist: lauded for its engineering prowess, but increasingly scrutinized for how that prowess is exercised.

Consumers occupy a particularly precarious position in many cases, and this ethical ecosystem is no exception to that. They are not merely passive recipients of product design, they're also participants, coerced into this cycle of disposability that few are aware of until it's too late and they're on their 2nd or 3rd pair of AirPods. They invest not only money but trust also into Apple's promise of quality. This trust is often betrayed when, after two or three years of use, those earbuds begin to fail, simply due to regular use and inevitable decay of battery performance. The issue with battery decay over time is not specific to this case, but the irreplaceability of those batteries is the central issue in AirPods. Consumers are not told, at point

³ Toh, Colin. "How Long Do AirPods Last (And How You Can Extend Their Lifespan)." Headphonesty, 1 Aug. 2023, www.headphonesty.com/2022/08/how-long-airpods-last/

of sale, that their product is functionally time-locked. The omission of this information is not incidental, but rather essential to the business model.

Independent repair professionals are another crucial group/stakeholder: one group which is often not properly considered in the design of products, such as AirPods. These are skilled technicians who, with non-AirPods, might be able to extend the lives of millions of devices. But with AirPods (and many other products), they are locked out -- not by lack of skill, but by deliberate obfuscation. Apple's refusal to provide parts, manuals, or even schematics for AirPods sends a clear message: this product is not meant to be touched or repaired by anyone but the original manufacturer. This gatekeeping doesn't just harm the livelihoods of repair professionals; it erodes an entire culture of repair and consumer independence.

Environmental advocates, too, find themselves entangled in this case. To them, AirPods are not merely gadgets, but rather they are emblems of a deeper systemic failure in how we design and dispose of technology. These devices, compact as they may be, contain components that are difficult to recycle safely. Lithium-ion batteries, when sealed inside plastic casings and glued in place, become ticking bombs in the e-waste stream. Some may argue that such design practices are symptoms of a broader disregard for environmental responsibility.

Finally, there are the engineers themselves, often caught between the ideals of technical elegance and the messier realities of ethical practice. Engineers face a troubling question: can innovation really, truly be called "progress" if it costs the loss of repairability, transparency, and sustainability? The AirPods case challenges engineers to reckon with the unintended consequences of their own ingenuity and to question whether the sleekest solution is really the right one for all parties.

Method

To analyze this case in its full ethical complexity, I will employ a hybrid ethical framework that draws from two crucial interwoven philosophies: virtue ethics and consequentialism. These lenses bring their own vocabulary and focus, allowing us to view the AirPods design not simply as a technical artifact but as an amalgamation of moral choices. This multidimensional approach recognizes that ethical design is not easily reducible to simple

metrics of efficiency or a dollar amount: it must also be interrogated in terms of character, consequence, and values.

Virtue ethics invites us to step back and ask what kind of moral character is revealed through this design. It centers not on the deontological abstract rules or the tallying of consequentialist consequences, but on the individual and collective moral disposition of those creating, marketing, and managing this product. When Apple engineers seal the AirPods battery inside a case impenetrable to all repairs, many must ask: what virtues or vices are present in that decision? Is it a demonstration of technical excellence, care, and responsibility? Is this a case of pride, myopic consideration, or a willful neglect of repair culture? This framework is especially pertinent and potent for corporate ethics, because it draws attention to the cultivated habits of organizations, questioning how they reason, what they prioritize, and how that internal culture shows up in products.

Consequentialism, in contrast, demands a more pragmatic look at the outcomes. It asks whether the benefits of a design decision -- such as improved aesthetics, sound quality, or market share -- truly outweigh its costs, especially when those costs are distributed across vulnerable ecosystems and global waste streams. From this perspective, we must consider: do the sleek aesthetics offered to the user justify their toxic legacy of millions of non-recyclable earbuds? Does the pleasure of seamless Bluetooth connectivity and impressive sound quality for their size outweigh the ethical burden of premature (and somewhat planned) obsolescence and e-waste? Consequentialism insists that ethical decisions and frameworks must be practical, scalable, and attentive to harm: especially when that harm, like battery leakage or landfill overflow, is invisible to most users but devastating over time.

By combining these two frameworks, I aim to provide an ethical analysis that is, all at once, principled, pragmatic, and design-oriented. Virtue ethics focuses on the character behind decisions and consequentialism on the taken actions and consequences. Together, they illuminate what is wrong with the current design of AirPods and what a more ethical and sustainable path forward might entail. This integrated approach resists moral reductionism and instead embraces the thorough, full complexity of technological and engineering ethics in a world where design decisions extend far beyond the boardroom or factory floor.

Ethical Issues

The design of AirPods presents a very dense and difficult thicket of ethical concerns, chief among them being the intentional and calculated practice of planned obsolescence. These earbuds are not simply hard to repair -- they are clearly engineered to resist it. The internal battery, which, as previously mentioned, inevitably degrades with time, is trapped in a shell that must be destroyed to be accessed. In doing so, the device enforces a life span that is artificially shortened. That limitation is not set by the limits of technology, but by the constraints imposed by design. Deliberately narrowing the product's usable life constitutes a calculated form of consumer manipulation, where users are nudged into unnecessary upgrades not because their device is broken, but because the system has been rigged to fail on schedule (with no repair options).

Equally troubling is the environmental aftermath of those design choices. AirPods are the very embodiment of modern e-waste: compact, composite, and practically impossible to recycle. Their small lithium-ion batteries, once expired, become hazardous materials that few recycling centers are equipped to handle. When disposed of improperly, a near inevitability given the scale of ownership and lack of infrastructure, they can leak toxic chemicals into the environment⁴. The ethical cost here is ecological degradation on a mass scale, hidden beneath a veneer of convenience and design.

Transparency, or the lack thereof, compounds the problem. Apple markets AirPods with the language of premium craftsmanship and responsible sourcing but fails to transparently inform customers of the product's unrecoverable, unrepairable battery or their limited lifespan. Ethical design demands openness, especially when user rights and environmental outcomes are on the line. When that transparency is withheld, it's not just bad communication, but rather a breach of trust.

Another major ethical infraction lies in the violation and degradation of user autonomy. When users buy AirPods, they believe they are purchasing a durable product, something they can

⁴ Institute for Energy Research. "The Environmental Impact of Lithium Batteries." Institute for Energy Research, 12 Nov. 2020, <https://www.instituteforenergyresearch.org/renewable/the-environmental-impact-of-lithium-batteries/>.

own, maintain, and perhaps pass on. The product is, however, designed not to be maintained. It cannot be upgraded or repaired. The user's choices are fixed: either replace the product entirely or tolerate its decay and the inconvenience of low battery life. Such a restriction of agency is deeply at odds with values of ownership -- a principle that lies at the heart of ethical consumption.

Finally, we must confront the systematic exclusion of third-party repair networks. This not only harms the livelihoods of these professionals and local economies, but it also consolidates control over the product lifecycle into the hands of the manufacturer. The result is a pseudo-monopolistic environment where repair becomes a luxury rather than a right. This means an economic system where ethical engineering is sacrificed at the altar of corporate control.

Stakeholder Perspectives

My robust and thorough ethical analysis must account for the diverse and often conflicting perspectives of those affected by a product's design. In the case of Apple's AirPods, the ethical terrain includes not only the corporation that manufactures them, but also the users who buy them, the professionals who attempt to repair them, and the environmental advocates who consider and trace their afterlife in landfills.

From Apple's point of view, the AirPods' seamless, sealed design is not merely a stylistic choice but an engineering necessity. The company argues that miniaturization is essential for performance: smaller components enable better sound fidelity, improved battery life, and increased portability. Additionally, sealing the product supports water resistance and contributes to the sleek, unified aesthetic that defines the Apple brand as we know it. On their website, Apple likes to gesture toward its corporate responsibility initiatives: emphasizing its recycling partnerships, trade-in programs, and commitments to carbon neutrality⁵. However, one must consider that these initiatives, while valuable and laudable, function largely downstream from the actual design process and do very little to address the core issue of repairability.

Consumers, particularly those unaware of the product's limitations at the point of sale, often experience frustration when they discover that even basic repairs or battery replacements

⁵Apple Inc. "Environment." Apple, 2025, www.apple.com/environment/

are impossible. For many, especially students, low-income users, or those living in regions without easy access to e-waste recycling, the inability to extend the life of a \$250 product represents not only an economic burden but a huge breach of consumer autonomy. The very notion of ownership is subtly undermined when a user cannot meaningfully maintain or modify the object they have purchased.

Professionals in electronics repair are often left with two options: damage the product beyond proper repair while attempting to open it or refuse the job altogether. Apple's design choices, namely using copious glue, proprietary screws, or fragile internal structures, effectively exclude skilled technicians from participating in the product's lifecycle. This not only impacts livelihoods but represents a larger moral failure: an ecosystem in which professional expertise is systematically devalued in the name of corporate control. Through the lens of virtue ethics, such choices raise concerns about pride, secrecy, and a lack of humility in design.

From an ecological standpoint, AirPods are a textbook example of planned obsolescence. Once the battery reaches the end of its short life, the entire device becomes functionally useless and often unrecyclable. Environmental groups may argue that repairability and product longevity are not optional ideals but foundational elements of any serious sustainability effort. Through the lens of consequentialist ethics, this design fails to meet the basic criterion of minimizing harm and maximizing good consequences of a decision.

Taken together, these stakeholder perspectives reveal the AirPods as a case study in ethical dissonance: where the priorities of design excellence and commercial success come into direct tension with the needs of users, workers, and the planet. A more ethically robust design process would not attempt to silence these perspectives but actively integrate them from the outset.

Option Evaluation

When confronting the ethics of AirPods' design, several paths present themselves, each with its own constellation of trade-offs. The first and least disruptive is to maintain the status quo: continuing with sealed designs while improving recycling pipelines and trade-in programs. From a business standpoint, this option is attractive. It preserves the visual and functional

minimalism that Apple has made iconic and demands no radical shifts in manufacturing. From an ethical lens, though it is an abdication of responsibility, it also has a positive impact or at least diminishes the current negative impact of these designs, thus this choice does net good. It leaves the deeper systemic problems unchallenged and unchanged. The entrenched cycle of obsolescence and the sidelining of repair culture both remain, but with a slightly less harmful consequence. This approach may also be seen as a form of ethical complacency, where a company's primary focus is on maintaining market dominance rather than cultivating a more sustainable and responsible approach to product design.

From a virtue ethics perspective, this option falls short of the moral ideals of integrity and responsibility. If the company or its engineers were to embrace a virtue of care, it would be more inclined to design products that could be more easily repaired, thus supporting a longer life cycle and reducing environmental harm. This path, while perhaps yielding some incremental positive change, does not fully embody the virtues that we would expect from a company of Apple's influence. It demonstrates a lack of moral courage when confronting the ethical dilemmas inherent in its business model.

A more transformative alternative lies in redesigning the product altogether. This would mean introducing a modular, repairable design. A design that sacrifices a few millimeters in size or elegance but allows repair. Batteries could be replaceable, and the housing could be openable. The trade-offs are very real: this approach may complicate water resistance, increase production costs, or unsettle the brand's visual identity. But the potential gains for the consumer are potentially profound.

From a consequentialist standpoint, this redesign offers the greatest good for the greatest number in the long term. It would reduce electronic waste, allow consumers to extend the life of their devices, and enable repair services to flourish. The long-term environmental and social benefits of such a change would far outweigh the immediate costs associated with altering manufacturing processes. It is worth noting that even though it will offer more good to consumers and other stakeholders, Apple does have to make some compromises, as previously mentioned. Those compromises could be justified, however, by the increasing trust and satisfaction from consumers and repair groups.

From a virtue ethics perspective, a modular design reflects virtues of responsibility, respect for users' autonomy, and care for the environment. By choosing to build products that are repairable, Apple would signal a commitment to moral integrity, signaling that the company values sustainability and user empowerment over its own image or short-term profits. Such a change would indicate a company that sees its role as not just a profit generator, but as a responsible player in a larger societal framework.

Between these poles there could exist a third main option: retaining the current design but vastly expanding repair infrastructure. Apple could offer battery replacement services through its stores, provide parts and tools to certified third parties, and educate consumers on the limitations and possibilities of their devices. This option presents a reasonable middle ground, enabling Apple to continue with its existing design while making incremental changes to support repairability and sustainability. This ideal compromise, however, unfortunately does not and cannot exist because the design of the AirPods is simply not conducive to repair. Apple could not effectively offer repair services at a reasonable price or capacity.

Recommendation

Given the analysis above, the most ethically robust course of action is clear: Apple should commit to a full redesign of AirPods that foregrounds modularity, repairability, and sustainability. This is not a rejection of innovation, but a call for a deeper, more mature understanding of what innovation and corporations ought to serve. A redesigned AirPods product line could stand not only as a feat of engineering, but as a beacon of responsible design.

Innovation and sustainability are not mutually exclusive, but rather they are mutually necessary. As engineers and designers, we must push past the tempting allure of the sealed box and imagine a future where form and function are just and complimentary. Repairability is not a regression to an outdated mode of consumption but should be seen and treated as a leap forward into a more humane, sustainable, and thoughtful future.

Conclusion

This case study highlights an uncomfortable truth: engineering excellence is sometimes divorced from ethical responsibility, which can produce products that are technically impressive but morally and ethically compromised. AirPods are a stunning achievement in many ways, but they are also, frankly, disposable by design.

Designing for repairability isn't just good for consumers -- it's a moral imperative for companies in an age of climate crisis, resource scarcity, and rising e-waste. Engineers cannot afford to be passive participants in this destructive cycle.

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